Tol muted (default cycle)



indigo	cyan	teal	green	olive	sand	rose	wine	purple	grey
#332288	#88CCEE	#44AA99	#117733	#999933	#DDCC77	#CC6677	#882255	#AA4499	#DDDDDD

Tol bright

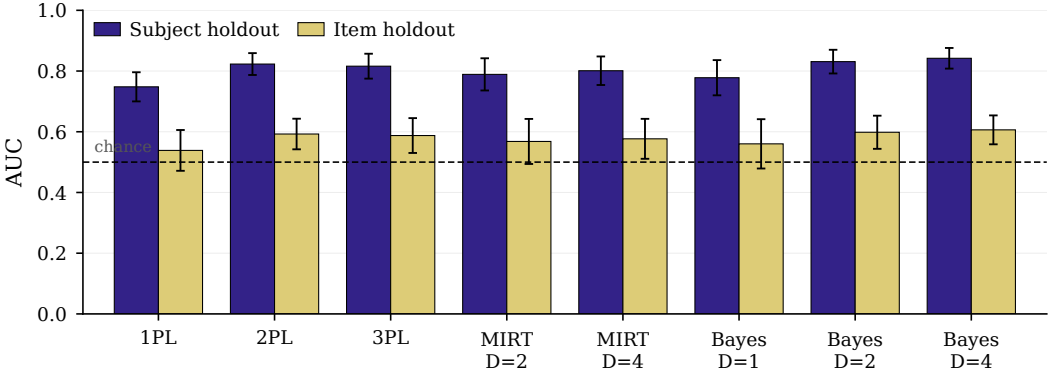


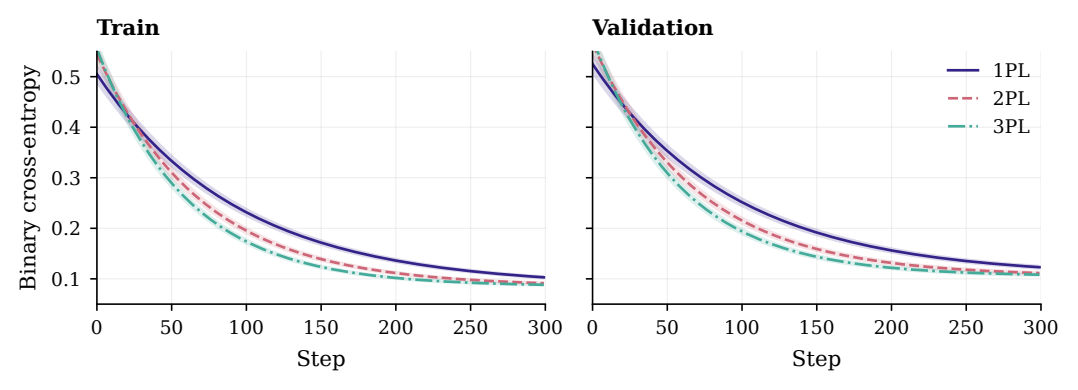
blue	cyan	green	yellow	red	purple	grey
#4477AA	#66CCEE	#228833	#CCBB44	#EE6677	#AA3377	#BBBBBB

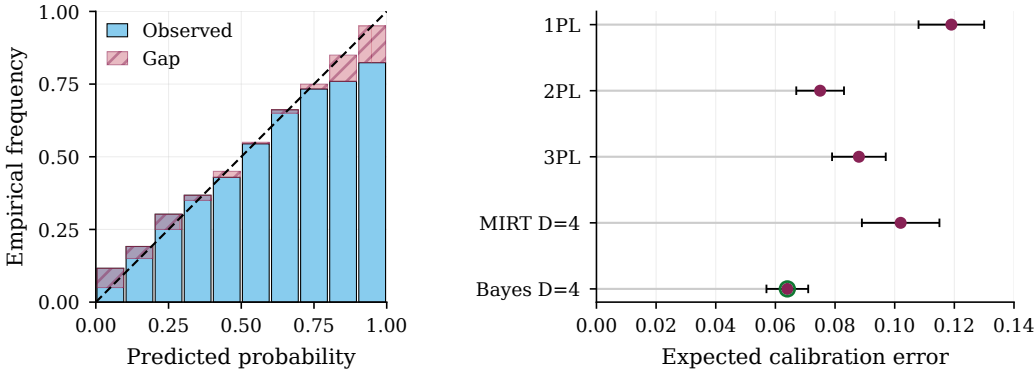
Tol high-contrast

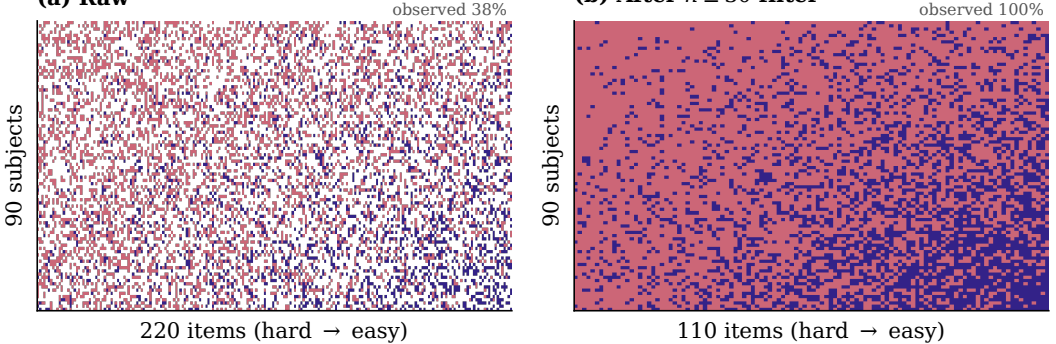


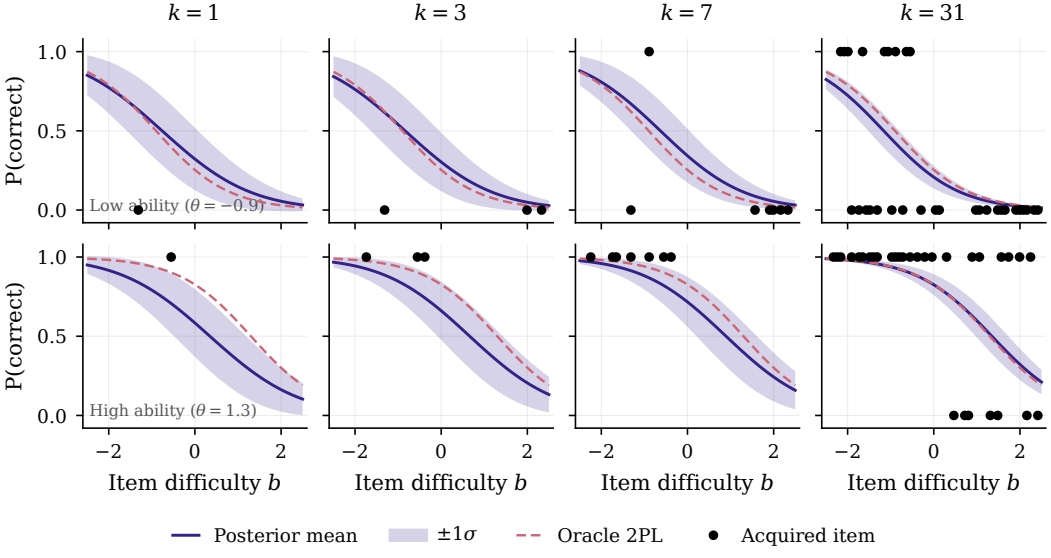
blue	yellow	red
#004488	#DDAA33	#BB5566

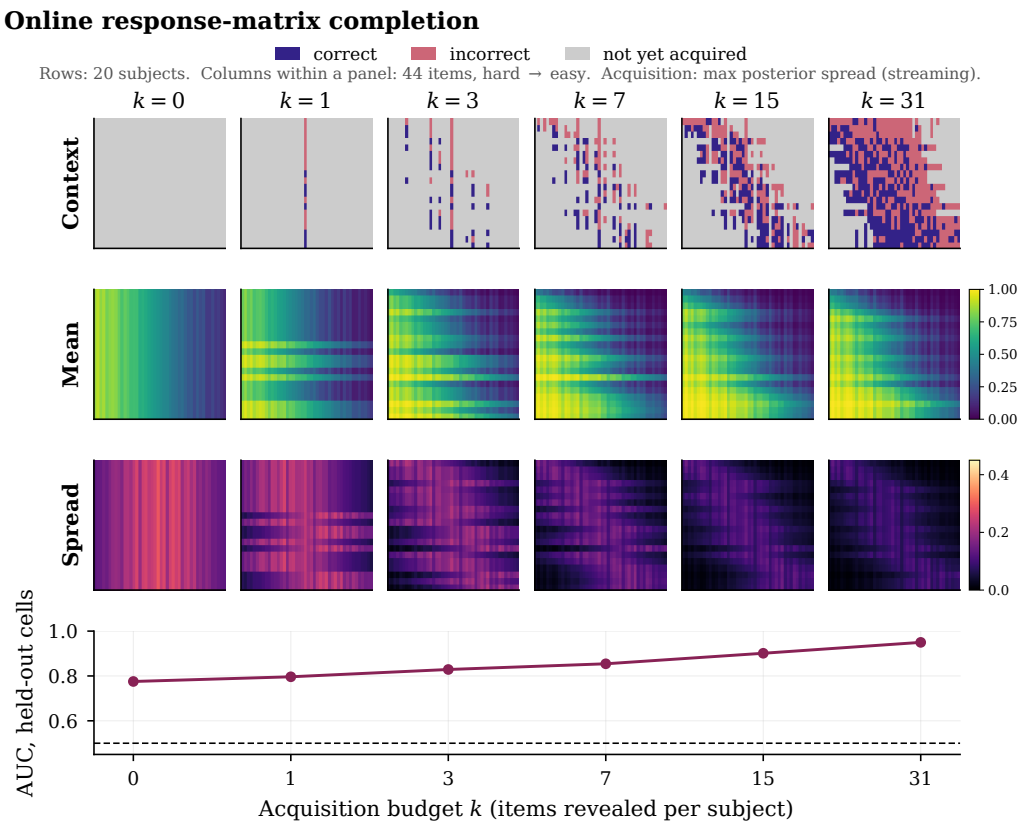

Subject holdout
Item holdout
AUC
1.0
0.8
0.6
0.4
0.2
0.0
chance
1PL
2PL
3PL
MIRT D=2
MIRT D=4
Bayes D=1
Bayes D=2
Bayes D=4

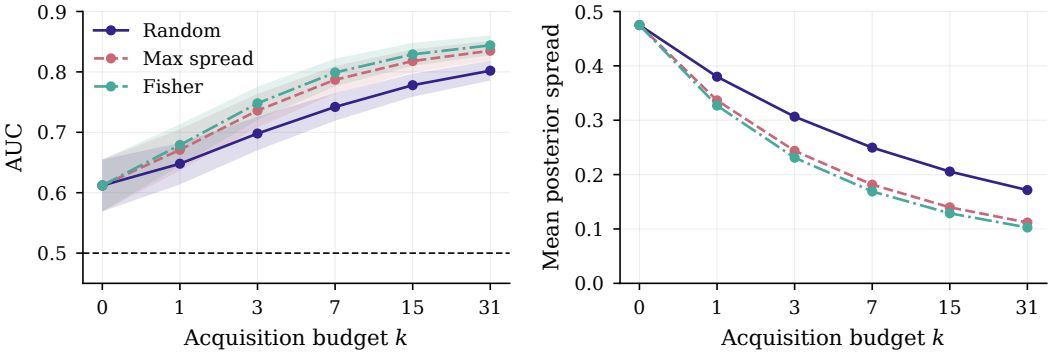

Train
Validation
Binary cross-entropy
0.5
0.4
0.3
0.2
0.1
0
50
100
150
200
250
300
Step
1PL
2PL
3PL


(a) Reliability, 2PL
Observed
Gap
Empirical frequency
Predicted probability
0.00
0.25
0.50
0.75
1.00
(b) ECE (lower is better)
1PL
2PL
3PL
MIRT D=4
Bayes D=4
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Expected calibration error


(a) Raw
observed 38%
90 subjects
220 items (hard → easy)
(b) After $n \geq 30$ filter
observed 100%
90 subjects
110 items (hard → easy)
correct
incorrect
unobserved


$k = 1$
$k = 3$
$k = 7$
$k = 31$
P(correct)
1.0
0.5
0.0
Low ability ($\theta = -0.9$)
High ability ($\theta = 1.3$)
−2
0
2
Item difficulty b
Posterior mean
$\pm 1\sigma$
Oracle 2PL
Acquired item


Online response-matrix completion
correct
incorrect
not yet acquired
Rows: 20 subjects. Columns within a panel: 44 items, hard → easy. Acquisition: max posterior spread (streaming).
k = 0
k = 1
k = 3
k = 7
k = 15
k = 31
Context
Mean
Spread
1.00
0.75
0.50
0.25
0.00
0.4
0.2
0.0
AUC, held-out cells
1.0
0.8
0.6
0
1
3
7
15
31
Acquisition budget k (items revealed per subject)


(a) Accuracy against budget
Random
Max spread
Fisher
AUC
0.9
0.8
0.7
0.6
0.5
0
1
3
7
15
31
Acquisition budget k
(b) Uncertainty against budget
Mean posterior spread
0.5
0.4
0.3
0.2
0.1
0.0
0
1
3
7
15
31
Acquisition budget k